

MAINTENANCE MANAGEMENT SYSTEM

Application Features, Access Guide & User Flow

Document Type	Application Documentation
Application	Maintenance Management System (MMS)
Technology Stack	Python (FastAPI) · HTML/CSS/JS · SQLite
Deployment	Local intranet — offline capable, no internet required
Status	Completed (v1)

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1. Application Overview

The Maintenance Management System (MMS) is an offline-capable, browser-based application designed for industrial plant operations. It allows maintenance teams to log, track, and manage equipment maintenance activities — both planned and conducted — in a structured, auditable format.

The application runs as a local intranet server on one Windows machine. All users on the same network access it through a standard web browser by navigating to the server's IP address. No software installation is required on users' computers.

Key Characteristics

- Fully offline — no internet required after installation
 - Multi-user access via web browser over the local network
 - Role-based access control (Administrator, Engineer / Operator, Viewer)
 - Full audit trail on all records — who created, edited, or deleted, with timestamps
 - File attachments (PDF, Excel, images) per maintenance record — up to 10 files
 - Structured CSV data entry and editing per record
 - PDF export of any maintenance record including embedded CSV data
 - Equipment hierarchy master list (Enterprise → Site → Area → Work Center → Work Unit → Equipment ID)
 - Soft-delete with Trash page and Administrator restore functionality
 - Bulk equipment upload via CSV file — up to 500 entries at once
-

2. User Roles & Permissions

The system has three roles. Access to features is enforced both in the UI and on every backend API endpoint – UI-level restrictions alone are not sufficient.

Permission	Administrator	Engineer / Operator	Viewer
View all maintenance records	Yes	Yes	Yes
View record details & attachments	Yes	Yes	Yes
Export record as PDF	Yes	Yes	Yes
Download attachments	Yes	Yes	Yes
Create new maintenance records	Yes	Yes	No
Edit own records	Yes	Yes	No
Edit any record	Yes	No	No
Upload / delete attachments on own records	Yes	Yes	No
Upload / edit CSV data on own records	Yes	Yes	No
Delete records (soft delete)	Yes	No	No
Bulk delete records	Yes	No	No
Restore deleted records	Yes	No	No
View Deleted Records (Trash) page	Yes	No	No
Manage equipment master list	Yes	No	No
Manage user accounts	Yes	No	No
Change own password	Yes	Yes	Yes

Note: Engineers may only edit or delete attachments on records they personally created. Administrators have no such restriction.

3. Access Guide

3a. Server Setup — IT / Administrator (One-Time Installation)

This section is for the person who installs and hosts the application on the Windows server or dedicated PC.

System Requirements

Requirement	Specification
Operating System	Windows 10 or Windows 11 (64-bit)
RAM	Minimum 2 GB recommended
Disk Space	Minimum 500 MB free (more depending on attachment volume)
Network	Connected to the local intranet (LAN or Wi-Fi)
Internet	Not required — the installer is fully offline
Privileges	Administrator rights required for installation

Step 1 — Run the Installer

1. Copy MMS_Setup_v1.0.exe to the server machine (e.g., USB drive or network share).
2. Double-click the installer. If prompted by Windows UAC, click Yes.
3. Follow the on-screen wizard. The installer automatically: installs Python 3.11, creates an isolated virtual environment, installs all Python dependencies from bundled offline wheel files, copies application files to C:\YKG_Maintenance\, initializes the SQLite database, and creates a Desktop shortcut and Start Menu entry.
4. When the wizard finishes, click Finish.

Step 2 — Launch the Application

5. Double-click the "Maintenance Management System" shortcut on the Desktop.
6. Wait 3–5 seconds. A background server process starts silently (no console window appears).
7. The default web browser opens automatically at <http://localhost:8000/index.html>, showing the Login page.

How the launcher works: The `launcher.pyw` script starts Uvicorn (a Python ASGI web server) on port 8000, binding to all network interfaces (0.0.0.0). It polls until the server is ready, then opens the browser. The server keeps running in the background as long as the launcher process is alive.

Step 3 — First-Time Login & Admin Setup

8. Log in with the default Administrator credentials: Username: admin / Password: AdminPass123!
9. Immediately go to Manage Users → click the avatar icon → change the default admin password.
10. Create user accounts for all staff members and assign appropriate roles.
11. Go to Equipment Master → add the plant's equipment hierarchy (or bulk upload via CSV).

Step 4 — Find the Server's IP Address

On the server machine, open Command Prompt and run: `ipconfig`

Look for the IPv4 Address under your active network adapter (e.g., 192.168.1.100). This is the address other users will type into their browser.

Step 5 — Keep the Server Running

The server runs as long as the launcher process is alive. For a dedicated server PC, leave the launcher running and consider configuring the application to start automatically at Windows startup using Task Scheduler or a startup shortcut.

The application does not require an active browser session on the server to serve other users — the server runs independently in the background.

3b. Client Access — Users Connecting via Browser

This section is for all plant staff who will use the system. No software installation is required on client devices.

What You Need

- A computer, laptop, or tablet connected to the same local network as the server
- A modern web browser: Chrome, Edge, or Firefox (recommended)

How to Access the Application

12. Open your web browser.
13. In the address bar, type the server's IP address and port — for example: `http://192.168.1.100:8000` (Replace with the actual IP address)
14. Press Enter. The MMS Login page will appear.
15. Enter your username and password (provided by your Administrator).
16. Click Login.

Tip: Bookmark the URL so you do not have to type it every time.

Note: The server must be running on the server machine for you to connect.

Session duration: 8 hours by default (one full shift). After 8 hours you will be automatically logged out.

4. Application Features Summary

Feature	Description
Maintenance Record Management	Create, view, edit, and soft-delete Planned and Conducted maintenance records
Equipment Hierarchy	Select equipment from a structured master list: Enterprise → Site → Area → Work Center → Work Unit → Equipment ID
File Attachments	Attach up to 10 files per record (PDF, Excel, JPG, PNG; max 10 MB each)
CSV Data Entry	Upload or manually enter tabular CSV data linked to a record; edit inline in a spreadsheet-style grid
PDF Export	Export any single record as a formatted PDF (includes record details, attachments list, and embedded CSV data)
Search, Filter & Sort	Search by equipment ID, responsible person, or remarks; filter by maintenance type; sort by any column ascending or descending
Audit Trail	Every record stores who created it, who last updated it, and (if deleted) who deleted it — all with timestamps
Soft Delete & Restore	Deleted records move to the Trash page (not permanently removed); Administrators can restore them at any time
Bulk Delete	Administrators can select multiple records on the Dashboard and delete them in one action (up to 100 at once)
User Management	Administrators create, deactivate, change roles, and reset passwords for user accounts
Role-Based Access	Three roles (Administrator, Engineer / Operator, Viewer) with enforced UI and API-level restrictions
Change Password	All users can change their own password via the sidebar avatar icon on any page
Equipment Bulk Upload	Administrators can upload a CSV file to add up to 500 equipment entries at once
Offline Operation	No internet required; all data is stored locally in an SQLite database on the host machine

5. Page-by-Page Flow & Explanation

Page 1 — Login (index.html)

URL: http://<server-ip>:8000/index.html or http://<server-ip>:8000/

Who sees it: Everyone — this is the application entry point

Layout

Sign In
Enter your credentials to continue

USERNAME *
Enter your username

PASSWORD *
Enter your password Show

Login

Intranet application — authorized personnel only

User Flow

17. User enters their username and password.
18. The Show / Hide button toggles password visibility.
19. On submit, credentials are sent to the authentication API (POST /api/auth/login).
20. On success, a JWT (JSON Web Token) is returned and stored in the browser's sessionStorage. The user is redirected to the Dashboard.
21. On failure (wrong credentials or inactive account), an inline error message is shown.

Key Behaviours

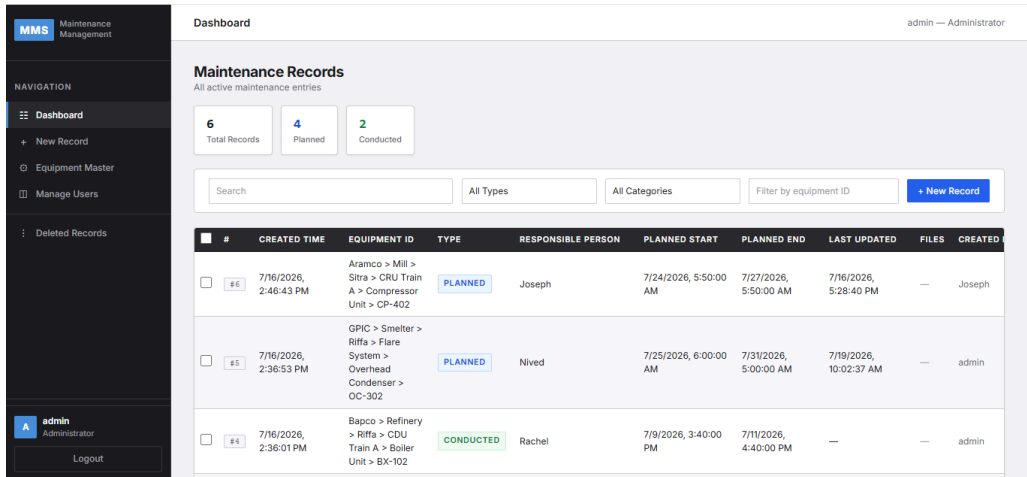
- If already logged in (valid token in sessionStorage), the page redirects directly to the Dashboard without showing the login form
- Both fields are validated as non-empty before the form is submitted
- The login token expires after 8 hours (configurable by the Administrator)

Page 2 — Dashboard (dashboard.html)

URL: http://<server-ip>:8000/dashboard.html

Who can access: All authenticated users (Administrator, Engineer / Operator, Viewer)

Layout



Sidebar Navigation Links

Link	Visible To
Dashboard	All roles
New Record	Administrator and Engineer / Operator only
Equipment Master	All roles
Manage Users	Administrator only
Deleted Records (Trash)	Administrator only

Dashboard Features

1. Summary Statistics Bar

Displays live counts: Total Records, Planned, and Conducted maintenance records.

2. Search, Filter & Sort Controls

- Search box: full-text search across Equipment ID, Equipment Full Path, Responsible Person, and Remarks
- Type filter: dropdown to show All / Planned / Conducted records
- Sort: click any column header to sort ascending or descending — click again to reverse

3. Records Table Columns

- Record ID, Maintenance Type badge (Planned / Conducted)
- Equipment ID and Equipment Full Path (Enterprise → Site → ... → Equipment)
- Created Time, Planned Start, Planned End
- Responsible Person
- Attachments count (e.g., "3 files")

4. Row Action Buttons

Button	Who Sees It	What It Does
View	All roles	Opens the Record Detail page for that record
Edit	Admin + Engineer (own records only)	Opens the form pre-filled with existing record data
Delete	Administrator only	Soft-deletes the record (moves to Trash)

5. Bulk Delete (Administrator only)

- A checkbox appears in each row for Administrators
- Select one or more records, then click Delete Selected
- A confirmation dialog appears before deletion
- Up to 100 records can be bulk-deleted at once

6. New Record Button

Prominent button in the page header — navigates to the record creation form (Engineers and Administrators only).

Page 3 — New / Edit Record Form (form.html)

URL (New): `http://<server-ip>:8000/form.html`

URL (Edit): `http://<server-ip>:8000/form.html?id=<record_id>`

Who can access: Create: Administrator, Engineer / Operator. Edit: Administrator (any record); Engineer (own records only)

Layout

The screenshot shows the 'New Maintenance Record' form. The left sidebar contains a navigation menu with 'New Record' highlighted. The main form area has a title 'New Maintenance Record' and a subtitle 'Complete all required fields marked with *'. The form is divided into several sections: 'CLASSIFICATION' with a 'MAINTENANCE TYPE' dropdown and an 'EQUIPMENT ID' search field; 'PERSONNEL' with a 'RESPONSIBLE PERSON' text field; 'PLANNED SCHEDULE' with 'PLANNED START' and 'PLANNED END' date pickers; and 'MAINTENANCE DETAILS' at the bottom. The user is logged in as 'admin - Administrator'.

Form Fields

Field	Required	Description
Maintenance Type	Yes	Dropdown — Planned or Conducted
Equipment	Yes	Searchable dropdown from the Equipment Master list — type to search; shows full hierarchy path
Responsible Person	Yes	Free-text; max 100 characters
Planned Start	No	Date and time picker
Planned End	No	Date and time picker — must be on or after Planned Start if both are provided
Operating Conditions	No	Text area; max 500 characters

Inventory / Consumables	No	Text area; max 500 characters
Remarks	No	Text area; max 2,000 characters
Attachments	No	Multi-file upload; accepted types: PDF, Excel (.xlsx, .xls), JPG, PNG; max 10 MB per file; max 10 files total per record

Equipment Selection

The equipment field is a live-search dropdown that queries the Equipment Master list. The user types part of the equipment ID or path, and matching entries appear in a dropdown. Only active equipment entries can be selected. The full hierarchy path is stored with the record (e.g., PlantCo > Main Site > Production Area > Line 1 > Unit A > EQ-001).

Edit Mode Behaviour

- The page loads the existing record data and pre-fills all fields
- Files already attached are listed below the attachment picker; new files can be added (total must not exceed 10)
- The Equipment field shows the currently assigned equipment but can be changed
- Submitting saves changes via the record update API

Validation

- Required fields are checked before submit
- Planned End cannot be before Planned Start
- File types and sizes are validated before upload
- Server-side validation returns specific error messages if any field fails
- A spinner / loading state is shown while the request is in progress

Page 4 — Record Detail (record-detail.html)

URL: http://<server-ip>:8000/record-detail.html?id=<record_id>

Who can access: All authenticated users (read-only for Viewers; editable for Administrators and Engineers)

Layout

MMS Maintenance Management

Record Details admin — Administrator

REC-6
CP-402 [Export PDF](#) [Edit Record](#)

IDENTIFICATION	
RECORD ID	6
CREATED TIME	7/16/2026, 2:46:43 PM
MAINTENANCE TYPE	Planned
EQUIPMENT ID	Aramco > Mill > Sitra > CRU Train A > Compressor Unit > CP-402
RESPONSIBLE PERSON	Joseph

AUDIT TRAIL	
CREATED BY	Joseph
CREATED DATE	2026-07-16T11:46:43.293080+00:00
UPDATED BY	admin
UPDATED DATE	2026-07-16T14:28:40.995309+00:00
LAST UPDATED	7/16/2026, 5:28:40 PM

ACTIVITY DETAILS	
OPERATING CONDITIONS	JFK
INVENTORY / CONSUMABLES	JFK
PLANNED START	7/24/2026, 5:50:00 AM
PLANNED END	7/27/2026, 5:50:00 AM

ATTACHMENT

—

admin
Administrator
[Logout](#)

Section A — Record Information Panel

Displays all record metadata in a structured layout:

- Record ID and Maintenance Type badge
- Equipment ID and full Equipment Path
- Created Time, Planned Start, Planned End, Last Updated Time
- Responsible Person, Operating Conditions, Inventory / Consumables, Remarks
- Audit information: Created By, Created Date, Updated By, Updated Date

Action buttons (role-dependent):

Button	Who Sees It	Action
Edit Record	Admin + Engineer (own records)	Opens form.html pre-filled with existing data
Export PDF	All roles	Downloads a formatted PDF of this record
Delete Record	Administrator only	Soft-deletes and returns to Dashboard

Section B — File Attachments Panel

Lists all files attached to the record in a table:

- Filename, file size, uploaded by, uploaded date
- Download button (all roles)
- Delete attachment button (Administrator or the Engineer who created the record)
- Engineers and Administrators can upload additional files directly from this page, subject to the 10-file-per-record limit

Section C — CSV Data Panel

This panel allows entry of structured tabular data (e.g., measurement logs, inspection readings) linked to the record.

State	What the User Sees
No CSV yet	An upload button to attach a CSV file
CSV uploaded	Data displayed in an inline editable spreadsheet grid
Editing	Cells become editable; rows can be added; Save and Undo buttons become active

- Upload a .csv file to populate the grid (max 5,000 rows, max 50 columns, max 10 MB)
- Edit any editable cell inline
- Add rows to the grid
- Save changes back to the server; Undo reverts to last saved state
- Download the current CSV data as a .csv file at any time

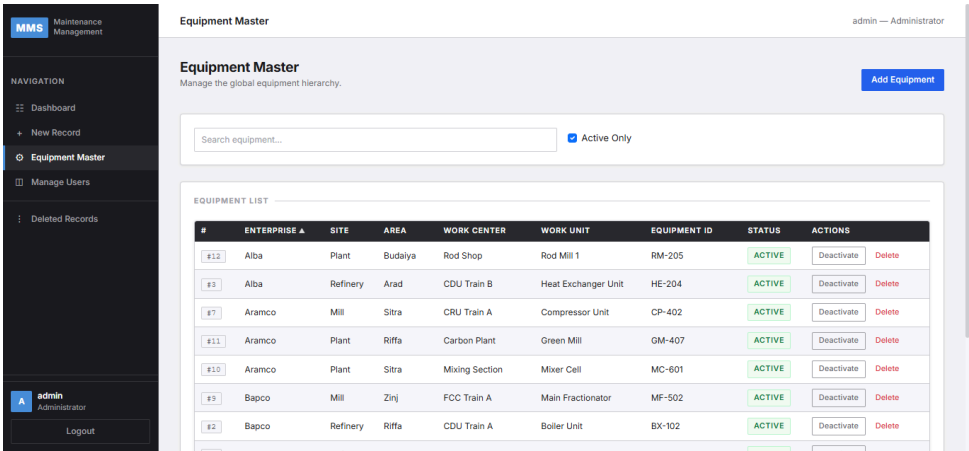
Page 5 — Equipment Master (equipment-master.html)

URL: http://<server-ip>:8000/equipment-master.html

View: All authenticated users

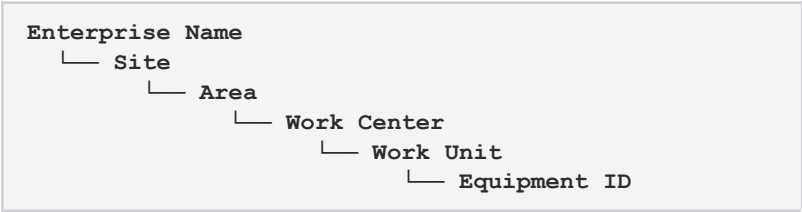
Add / Edit / Delete: Administrator only

Layout



Purpose

Maintains the master list of equipment that users can select when creating maintenance records. Equipment is organized in a six-level hierarchy:



Example: Bapco > Sitra Refinery > Crude Distillation Unit > CDU Train A > Boiler Unit > BX-101

Features

Equipment Table

- Displays all equipment entries (active and / or inactive, based on filter)
- Columns: Enterprise, Site, Area, Work Center, Work Unit, Equipment ID, Status
- Search by full path keyword

Add Equipment (Administrator only)

- A form with fields for all six hierarchy levels
- Duplicate entries (same full path) are rejected by the server

Edit Equipment (Administrator only)

- Supports toggling Active / Inactive status
- Deactivating equipment prevents it from being selected in new records, but does not affect existing records that already reference it

Delete Equipment (Administrator only)

- Permanently deletes the equipment entry
- Cannot delete equipment referenced by active maintenance records – the server returns an error; deactivate it instead

Bulk Upload (Administrator only)

- Upload a CSV file with up to 500 equipment entries
- Required CSV columns: Enterprise Name, Site, Area, Work Center, Work Unit, Equipment ID
- The file is validated before any data is inserted – if any row has errors, nothing is saved
- Duplicate entries are silently skipped (reported as a skipped count in the response)
- Maximum file size: 5 MB
- A CSV template file can be downloaded directly from this page

Page 6 – Manage Users (manage-users.html)

URL: <http://<server-ip>:8000/manage-users.html>

Who can access: Administrator only – Engineers and Viewers are redirected to the Dashboard

Layout

Manage Users

Create new user accounts. Administrator accounts are provisioned by the system owner only.

CREATE NEW USER

USERNAME *
Enter Name

ROLE *
— Select Role —

PASSWORD *
Min. 8 characters

CONFIRM PASSWORD *
Repeat password

Create User Cancel

EXISTING USERS

#	USERNAME	ROLE	STATUS	ACTIONS
#1	admin	Administrator	ACTIVE	(Your Account)
#2	Joseph	Engineer / Operator	ACTIVE	Deactivate Reset Password
#3	Ramesh	Viewer	ACTIVE	Deactivate Reset Password

Features

Users Table

- Displays all user accounts with: Username, Role, Status (Active / Inactive), and Action buttons

Create New User

- Opens a form with: Username (max 50 characters, must be unique), Password (minimum 8 characters), and Role
- A double confirmation step (type the new username to confirm) is required before the account is created
- On submit, the account is created immediately and appears in the table

Edit User

- Administrators can change: Role (e.g., promote a Viewer to Engineer / Operator), Status (Active ↔ Inactive – deactivating prevents login), and Password (set a temporary password for the user)

- Changing a user's role requires the user to log out and back in for the new permissions to take effect

Safety Rules

- An Administrator cannot deactivate their own account (prevents accidental logout)
- An Administrator cannot change their own role
- The Administrator role cannot be assigned via the role change UI — only Viewer and Engineer / Operator transitions are supported

Change Own Password

- All roles can change their own password by clicking the avatar icon in the sidebar footer
- Requires entering the current password plus the new password (minimum 8 characters)

Page 7 — Deleted Records / Trash (trash.html)

URL: <http://<server-ip>:8000/trash.html>

Who can access: Administrator only

Layout

#	EQUIPMENT ID	TYPE	RESPONSIBLE PERSON	CREATED BY	DELETED BY	DELETED DATE	ACTION
1	VT-301	Conducted	Vineeth	admin	admin	7/20/2026, 10:38:36 AM	Restore
2	MC-601	Planned	Rajiv	admin	admin	7/20/2026, 10:38:36 AM	Restore

Purpose

Acts as a safety net. Records that have been deleted from the Dashboard are moved here rather than being permanently erased. This allows recovery of accidentally deleted records.

How Soft-Delete Works

- When a record is deleted, the database row is NOT removed
- Instead, deleted_by (username) and deleted_date (timestamp) are stamped on the record
- The record disappears from the Dashboard but appears in this Trash page
- All attachments and CSV data associated with the record remain intact

Features

Deleted Records Table — displays:

- Record ID, Maintenance Type, Equipment ID, Created Time

- Responsible Person, Created By, Deleted By, Deleted Date

Restore Record

- Each row has a Restore button (Administrator only)
- A single confirmation dialog is shown before restoring
- Restoring clears the deleted_by and deleted_date fields, making the record active again and returning it to the main Dashboard immediately

No Permanent Delete from UI

- The application does not provide a permanent delete button in the UI
 - Data is never permanently removed through normal operation
 - To fully purge data, an Administrator must access the SQLite database file directly
-

6. Data Model Reference

Maintenance Record Fields

Field	Type	Description
id	Integer	Auto-assigned unique record ID
maintenance_type	Text	"Planned" or "Conducted"
created_time	Timestamp	When the maintenance activity was logged
equipment_id	Text	Equipment identifier (e.g., P-001)
equipment_full_path	Text	Full hierarchy path (e.g., PlantCo > Main Site > ...)
responsible_person	Text	Name of the person responsible for the maintenance activity
planned_start	Timestamp	Scheduled start date / time (optional)
planned_end	Timestamp	Scheduled end date / time (optional)
operating_conditions	Text	Description of operating conditions (optional, max 500 chars)
inventory_consumables	Text	Parts or consumables used (optional, max 500 chars)
remarks	Text	General notes (optional, max 2,000 chars)
last_updated_time	Timestamp	When the record was last modified
created_by	Text	Username of the user who created the record
created_date	Timestamp	Audit timestamp of row creation
updated_by	Text	Username of the last user to modify the record
updated_date	Timestamp	Audit timestamp of last modification
deleted_by	Text	Username who deleted it (null = not deleted)
deleted_date	Timestamp	When it was deleted (null = not deleted)

7. File Attachment Rules

Rule	Value
Maximum files per record	10 (combined across create and all edits)
Maximum file size	10 MB per file (configurable via server environment variable)
Allowed file types	.pdf, .xlsx, .xls, .jpg, .jpeg, .png
Storage location	C:\YKG_Maintenance\backend\storage\attachments\
File naming on disk	UUID-based (e.g., a3f1c8...) — original filename preserved in database
Delete behaviour	Deleting an attachment removes both the database row and the physical file
Validation	Both file extension AND MIME type are checked server-side before any file is saved

8. CSV Data Feature

Each maintenance record can have one CSV dataset attached to it. This is designed for tabular data such as sensor readings, inspection measurement logs, test results tables, and calibration data.

Capability	Detail
Upload	Upload a .csv file from the Record Detail page. The file is parsed in memory and stored as JSON in the database — not as a file on disk.
Edit	After uploading, the data is shown in an inline spreadsheet grid where cells and rows can be edited directly in the browser.
Save	The Save Changes button writes the current grid state back to the database. Undo reverts to the last saved state.
Download	The current CSV data can be downloaded as a .csv file at any time.
PDF Export	When exporting a record as PDF, the CSV data (up to 500 rows) is embedded in the PDF as a formatted table.

Limits

Limit	Value
Maximum columns	50
Maximum rows	5,000
Maximum cell content	1,000 characters per cell
Maximum upload file size	10 MB
Rows embedded in PDF export	Up to 500 rows (truncated with a note if the dataset is larger)

9. PDF Export Feature

Any authenticated user (all roles) can export a maintenance record as a professionally formatted PDF document.

Trigger: Click the Export PDF button on the Record Detail page

PDF Contents

22. Document header — Record ID, Equipment ID, exported by, export timestamp
23. Record Details table — all record fields (maintenance type, equipment ID, timestamps, responsible person, conditions, consumables, remarks, and audit information)
24. Attachments list — filename, uploaded by, and upload date (only if attachments exist)
25. CSV Data table — embedded tabular CSV data (only if CSV data exists; truncated to 500 rows for very large datasets with a note)

Technical Notes

- Generated entirely in memory using the reportlab library — no temporary files are written to disk
 - Delivered as a streaming download named record_<id>_export.pdf
 - The PDF uses a clean, professional A4 layout with alternating row colours and a dark header style
 - The PDF reflects the record data at the moment of export
-

10. Security Model

Mechanism	Detail
Authentication	JWT (JSON Web Token) — signed with HS256, stored in browser sessionStorage
Token Expiry	8 hours by default (configurable via ACCESS_TOKEN_EXPIRE_MINUTES in the server .env file)
Password Hashing	bcrypt via passlib — passwords are never stored in plain text
Role Enforcement	Every API endpoint checks the user's role from the JWT — UI restrictions alone are not sufficient and are not relied upon for security
Session Storage	Token is stored in sessionStorage (cleared when the browser tab is closed, unlike localStorage)
File Validation	File extension AND MIME type are both checked server-side before any file is saved to disk
SQL Injection Prevention	All database queries use parameterized statements — no string concatenation for user input
CORS	Configurable via ALLOWED_ORIGINS in the server .env file; defaults to allowing all origins on the local intranet

11. Uninstallation & Data Preservation

To Uninstall

26. Open Windows Settings → Apps & Features (or Add or remove programs).
27. Find Maintenance Management System and click Uninstall.
28. The uninstaller will: stop any running server or launcher processes, remove application files, remove the Python virtual environment (venv/), and remove the bundled wheel cache (wheels/).

Data is intentionally preserved: The database (C:\YKG_Maintenance\backend\storage\maintenance.db) and all uploaded attachment files (C:\YKG_Maintenance\backend\storage\attachments\) are NOT deleted by the uninstaller. This prevents accidental data loss.

To completely remove all data, manually delete the entire C:\YKG_Maintenance\ folder after running the uninstaller.
